

Business Problem Statement

Background

The business collects transactional data — demographics, purchases, discounts, subscriptions, and shipping — but has never systematically analyzed it. Decisions on marketing, pricing, and retention are based on assumption rather than evidence.

Problem

There's no clear visibility into **who the customers are, what drives their spending, and which strategies (discounts, subscriptions, shipping) actually move revenue**. This risks inefficient marketing spend, avoidable margin loss from discounting, and missed retention opportunities.

Objectives

- Identify which customer segments and products drive the most revenue
- Determine whether discounts justify their cost in lost margin
- Evaluate whether the subscription program adds real value
- Segment customers (New / Returning / Loyal) to target retention
- Support inventory and marketing decisions with data, not guesswork

Key Questions

1. Revenue: male vs. female customers?
2. Discount users spending above average?
3. Top 5 products by review rating?
4. Standard vs. Express shipping — purchase amount comparison?
5. Subscribers vs. non-subscribers — spend and revenue?
6. Top 5 products by discount usage rate?
7. Customer segmentation: New / Returning / Loyal counts?
8. Top 3 products per category?
9. Do repeat buyers subscribe more?
10. Revenue by age group?

Scope

In: 3,900-record transaction dataset, cleaning, SQL analysis, Power BI dashboard **Out:** real-time data, predictive modeling, PII-level analysis

Deliverables

- Cleaned dataset in PostgreSQL
- SQL answers to all 10 questions
- Interactive Power BI dashboard
- Actionable business recommendations